<u>2009 H2 Biology Preliminary Examination Paper 3 SMS</u>

<u>Question 1 [15m]</u>

(a) State the name of the test that was carried out. [1]

- **Paternity Test/ DNA Fingerprinting / Genetic Fingerprinting**

Common error: gel electrophoresis

(b)(i) Explain the purpose of the DNA ladder in Fig. 1.1. [2]

- **Each band serves as a <u>comparison / reference</u>**
- **for the <u>size / no. of repeats</u> present / found in the individuals tested.**

(b)(ii) DNA is colourless.
State how the DNA bands in the gel can be made visible. [1]

- **Ethidium bromide under UV light OR**
- **staining with methylene blue**

(c) With reference to Fig. 1.1, explain why
(i) some bands in a lane are higher than others; [4]

- **Higher bands represent the DNA fragments with a larger number of repeated units**
- **The fragments of DNA with a larger number of repeated units are longer**
- **The rate of movement in a given time depends on the size of the DNA fragment**
- **The longer a DNA fragment, the harder it is for that fragment to travel through the pores in the gel, and so it migrates slower than the smaller fragments with fewer repeats.**

(c)(ii) the lane for Dad#3 has only one band; [2]

- **They are homozygous at this locus.**
- **This means that they have the same number of repeats on both homologous chromosome 13**
- **For example: Dad #3 has 25 repeats on both versions of his chromosome 13.**

(c)(iii) some lanes have two bands. [2]

- **They are heterozygous at this locus.**
- **This means that they have a different number of repeats on each homologue of chromosome 13.**

(d) (i) With reference to Fig. 1.1, explain if the paternity of Baby A is conclusive. [2]

- **Baby A has a band in n=25 and there is only one set of parents that could have given a chromosome 13 with n=25 to their offspring, couple #3.**
- **This is because Dad #3 is the <u>only parent with n=25 @ chromosome</u>**

(d)(ii) Suggest an alternative way by which DNA from all individuals can be analyzed to confirm the paternity of Baby A. [1]

- **Analysis of RFLP, other short tandem repeats (STRs).**

<u>Question 2 [20m]</u>

(a)(i) Describe what is meant by a *gene mutation*. [2]

- **change in sequence of base pairs (in a DNA molecule) ;**
- **details of gene mutation; e.g. addition / substitution / deletion / frame shift mutation**
- **altered DNA sequence may code for a different protein / may code for no protein**

(a)(ii) Explain how CF is inherited. [3]

- **Cystic fibrosis is a <u>recessive</u> genetic disorder**
- **Caused by mutant recessive allele located on chromosome 7 / an autosome;**
- **homozygote recessive individuals = suffers from cystic fibrosis / is affected by the condition ;**
- **A heterozygote = carrier of one copy of the recessive allele;**
- **Both parents of sufferer must be carriers;**
- **1 in 4 chance of offspring being affected (when both parents are carriers)**

@1/2 m each

Common errors: (1) ref to CF as a sex-linked condition and (2) description of mutation leading to lack of CFTR protein leading to symptoms of CF.

(a)(iii) Describe briefly **two** symptoms of CF. [2]

Any 2 symptoms:

- **thick / dehydrated, mucus builds up in lungs and gut;**
- **blocks up the airways and becomes a breeding ground for bacteria → bacterial infections in lungs ;**
- **results in severe cough, wheezing, blocked sinuses and difficulty in breathing.**
- **The chest may develop into a barrel-shape / scar / damage lungs due to increased effort to breathe properly**

- **mucus blocks secretion of digestive enzymes (from pancreas) ;**
- **Digestion of food is thus impaired and the CF-sufferer may become malnourished inadequate, digestion / absorption ;**

- **Baby with CF: large amounts of very salty sweat produced as the sweat dries the baby becomes coated with a white deposit of salt.**

- **mucus blocks sperm duct / males sterile ;**

@ 1/2 m each

(b) With reference to the information given, explain why some mutant CFTR alleles allow normal digestive functioning of the pancreas and others do not. [3]

- **mutation may give different amino acid / primary structure ;**

A ref to stop codon → results in premature termination of polypeptide synthesis / truncated polypeptide formed → nonfunctional CFTR protein

- **some mutations alter molecular shape / tertiary structure / binding of CFTR protein;**
- **so unable to, accept / transport HCO_3^- ;**
- **so increase in acidity / decrease in pH of tissue secretion;**
- **effect on pancreatic enzyme(s) → unable to carry out transport of HCO_3^-;**
- **ref pH optimum of pancreatic enzyme(s) → function normally at alkaline pH ;**
- **poor digestion of protein / lipid / starch ;**
- **some mutations (ref to mutations 6 – 10), allow some transport of HCO_3^- (between 33 – 44% HCO_3^- transported);**
- **33% HCO_3^- transported allows normal digestive function / < 6% HCO_3^- transported (ref to mutations 1 – 5) does not allow normal digestive function ;**

(c) With reference to Table 2.2, explain the role of the glucose-sensitive promoter in this gene therapy. [2]

- **insulin is a polypeptide / protein (hormone) ;**
- **attachment of RNA polymerase to glucose-sensitive promoter region, switches on transcription of insulin gene;**
- **resulting in production of insulin ;**
- **as mean blood glucose concentration rises from 100 to 500 mg dm^{-3}, insulin production increases from 0.3 t0 7.0 ng cm^{-3};**
- **insulin only produced when needed ;**
- **ref to allowing homeostasis / negative feedback ;**

@ ½ m each

(d) With reference to Fig. 2.1, discuss the possible **benefits** and **problems** of using this gene therapy in the treatment of diabetes in humans, rather than taking insulin. [4]

<u>Benefits</u>

- **avoids injections / pain of injections / children's fear of injections ;**

Common error: reference to <u>oral intake</u> of insulin

- **mimics normal pancreatic behaviour ;**
- **more stable homeostasis / reduced highs and lows in blood sugar (reference to mean blood glucose concentration rising above the norm [+150 mg dm^{-3} above the norm] above glucose meal & reduced / returned <u>back to norm</u> after 6 hours);**
- **less chance of hypoglycaemia / hyperglycaemia ;**
- **less restriction on lifestyle ;**
- **no need to measure blood sugar ;**

max 2

<u>Problems</u>

- **rejection ;**
- **cells could lodge elsewhere ;**
- **may take longer to act (since time is required for production of insulin in treated patients] ;**
- **AVP ; e.g. rat data may not be applicable to humans,**
- **transgene may have unforeseen effect**

max 2

(e) As the result of the Human Genome Project, scientists have the ability to test for many inherited diseases.

State **two** advantages and **two** disadvantages of the ability to test for inherited diseases. [4]

<u>Advantages (Any 2)</u>

- **Able to screen embryo / diagnose disease early / before symptoms show;**
- **Treat person / disease, early / before symptoms appear / AW;**
- **Chance to carry out gene therapy;**

Reject: ref to genetic engineering / designer babies, etc.

- **Allows termination / abortion / AW;**
- **Fewer people will have that disease;**
- **Informed decision making for to-be parents (e.g. about whether to have children);**
- **Enables change in lifestyle / take preventive action (to delay onset of disease) ;**

<u>disadvantages (Any 2)</u>

- **genetic underclass / ref to prejudice / stigmatised by society's attitude towards genetic differences; AW**
- **discrimination / abuse of knowledge, by employers (*NOT employees*) / insurance companies / banks / medical profession e.g. employer may not offer that person a job or insurance may be refused;**

Reject: ref to economic factors

- **ref. to privacy of information ;**
- **inaccuracy of testing;**
- **allow termination / abortion / AW ;**
- **discourages people from having children ;**
- **AVP e.g. increased number of terminations / moral / psychological / social issues**

<u>Question 3 [10m]</u>

(a) One mature plant grows from each glyphosate -resistant plant cell. Explain why all the cells of the mature plant contain the gene for glyphosate – resistance. [1]

- **The cells are <u>genetically identical / clone</u> of the glyphosate cells due to <u>mitotic division</u> of the cell.**

(b) Outline the technique of tissue culture needed after obtaining matured plants from the transformed cells which are resistant to the herbicide. [3]

- **The explant is first sterilized using <u>sodium hypochlorite</u>**
- **The explant is then transferred to an <u>aseptic culture medium</u> that contains <u>nutrients, growth factors/hormones</u>.**
- **The cells of the explant will divide by <u>mitosis</u> to form a <u>callus</u>**
- **The callus can be <u>subcultured</u> into many calli in different culture**
- **The <u>ratio of plant hormones</u> will result in the plant tissues developing into <u>specialized tissues</u>.**
- **e.g. of a ratio leading to either root inducing or shoot inducing**
- **When a <u>plantlet is formed</u>, it can be transferred to <u>soil</u> to grow into a whole plant.**

(c) Suggest and explain one possible problem associated with the herbicides together with genetically modified herbicide-resistant crop plants. [1]

- **The creation of <u>superweeds</u> with <u>horizontal gene transfer</u>**
- **The <u>production of toxins</u> by the gene that can cause <u>allergy</u> in human beings**

(d) With reference to the figure,

(i) explain what is meant by haploid callus. [2]

- **It refers to <u>unspecialized cells</u> that are <u>actively dividing by</u> <u>mitosis</u>. The cells will only have <u>one set of chromosomes</u> from <u>one parent</u>.**

(ii) state one precaution that would be taken when transferring material onto culture medium 1 or 2. [1]

- **Ensure that the environment for the transfer is aseptic to prevent contamination.**

(iii) suggest why variation may be generated in plantlets derived from the same callus. [1]

- **It can be due to somaclonal variation or chromosomal mutations.**

(iv) Suggest how colchicine could bring about the doubling of chromosome number in haploid cells. [1]

- **It may prevent the sister chromatids from separating during anaphase.**

Accept logical answers as well.

Question 4 [20m]

(a) Using a **named example**, describe the characteristics of adult stem cells. [8]

- **Adult stem cells are unspecialized/undifferentiated cells that are actively dividing by mitosis to replace worn out cells/injuries/ cell death through diseases to maintain the tissue that they are found in.**
- **The mitotic division also ensures that there is a pool of stem cells (self renewal). The stem cells are limited in quantities and can only be found in specific areas of the body.**
- **One example will be the haemopietic stem cells that can only be found in the bone marrow.**
- **The haemopietic stem cells will constantly divide to replace cells such as the red blood cells that are worn out in three to four months.**
- **The stem cells will take cues from the cells surrounding them to differentiate into specialized cells or tissues.**
- **Adult stem are multipotent, they are able to differentiate into the cell types of the tissue that the stem cells are located in and not all cell types, and for example haemopioetic cells are only able to differentiate into cells such as red blood cells and white blood cells.**

(b) Explain why embryonic stem cells are preferred over adult stem cells for the treatment of genetic diseases. [6]

- **Embryonic stem cells are totipotent / pluripotent and it can take cues from surrounding tissues to differentiate into all cell types that can be found in the body, whereas adult stems cells are multipotent.**

- **Embryonic stem cells can be found in high quantities from infertility treatments. In addition, embryonic stem cells have longer lifespan and can be cultured for a long period in laboratories to result in a large supply of stem cells for treatments.**

- **Adult stem cells can only be found in limited quantities in certain tissues and they can be difficult to source for and hard to extract.**

- **Adult stem cells have to be retrieved from the patient's own tissues which may endanger the life of the patient whereas embryonic stem cells can be obtained from cell cultures that will not endanger the patient's life.**

(c) Using named examples, suggest three advantages of the use biotechnology in the enhancement of food quality and quantity. [6]

Any three from below

- **Genetic engineering of plants such as the insertion of Bt delta gene from the bacteria *Bacillus thuringiensis*; it can kill larvae of the *Lepidoptera* species. Thus with less pests, the quantity of BT plants can be increased.**

- **The insertion of genes that can increase plant's tolerance to abiotic stresses (e.g. salinity and drought) will result in the farming of land that is not suitable for agriculture, this increases the quantity of food.**

- **The insertion of genes into animals; such as the insertion of genes encoding for growth hormones into salmon can result in higher / faster growth in the fish. This results in more production of meat in a shorter time. This increases the quantity of food.**

- **The insertion of genes that encodes for glycophosate resistance in plants such as cotton; Round up ready plants will result in the plant being resistant to herbicides. Thus when glycophosate based herbicides are used, weeds are killed while the plants are not. This will increase the resources for Round up ready plants to grow, increasing the quantity of food.**

THE END